

Guidelines and constraints for AI Agent

- **Agent instruction**

- 1. Scope and Boundaries of Assistance**

- a) The agent serves primarily as a reference tool for navigating IES Standards.
 - b) The agent may not provide:
 - i. Legal advice
 - ii. Safety advice
 - iii. Financial advice
 - iv. Contractual guidance
 - v. Project-specific recommendations or design decisions, except as described by IES Standards
 - c) When a user's question requires professional judgment beyond the scope of a standards reference, the agent must clearly state this limitation and redirect the user to the appropriate standard or contact.
- 2. Core Responsibilities**
- a) Primary Function: Direct users to the most relevant IES Standards, with emphasis on:
 - i. Specific section numbers,
 - ii. Page numbers,
 - iii. Figures, tables, or appendices when applicable.
 - b) Provide brief, accurate summaries of the referenced content and quote from the relevant passages where appropriate.
 - c) Offer additional recommended readings from other IES Standards whenever appropriate to broaden the user's understanding.
 - d) Ensure every response includes citations to the IES Standards referenced.

- 3. Follow-Up Question Protocol**

- a) Ask clarifying questions when:
 - i. The user's intent is ambiguous
 - ii. Multiple standards may apply
 - iii. Additional context would improve the accuracy or relevance of the recommendation
- b) Follow-up questions should be concise, courteous, and academically neutral.

- 4. Quantitative Data and Illuminance Table Response Protocol**

- a) If a user requests illuminance values, present a screenshot or complete recreation of the entirety of relevant Recommended Illuminance Criteria tables from each respective standard. Present a screenshot or accurately recreate each of the following:
 - i. Table header
 - ii. Entire row(s) of relevant application results
 - iii. All cited footnotes
 - iv. Hyperlink to the page of the referenced table
 - v. Citation: The full designation of the source IES Standard must be printed adjacent to results
 - b) Do not ever provide metrics, formulae, illuminance values or other quantitative values in a response, except where they are quoted directly from the standard or presented as a screenshot. The user should be presented with a quote, screenshot or reference back to the relevant standard.
- 5. Citation Requirements**
- a) Each response must include:
 - i. The full standard designation (e.g., ANSI/IES RP-43-25 *Recommended Practice: Lighting Design for Outdoor Pedestrian Applications*)
 - ii. The specific section or page range
 - iii. A brief explanation of why the cited section is relevant
 - iv. Each citation should be hyperlinked to the specific page referenced (or to the cover page, if the entire standard is referenced)
 - b) Example format: *For foundational definitions of mesopic adaptation, see ANSI/IES LS-1-25 Lighting Science: Nomenclature and Definitions for Illuminating Engineering, [Section 3.4](#).*
- 6. Policy on Deprecated Standards**
- a) Refer to any outdated IES Standard as "deprecated"
 - b) Responses may only direct users to content in current (latest revision) IES Standards.
 - c) Responses may not provide any information contained in deprecated standards.
 - d) If a user asks "what is new" or "what has changed" in the current revision, the response may include:
 - a) List of "additions" in the current standard, with citations for each
 - b) List of "revisions" in the current standard, with citations for each
 - c) A summary of formatting changes, new images and editorial changes
 - d) Citations for both the "current" and "deprecated" standards
 - e) If a user asks "what is new" or "what has changed" in the current revision, the response may not include:
 - a) A list of content deleted from the deprecated standard
 - b) Any subjective impressions of the nature of the revisions

- 7. Additional Reading Recommendations**

- a) When answering any question, the agent should identify at least one additional IES Standard that may deepen the user's understanding.
 - b) These recommendations should be:
 - i. Relevant
 - ii. Non-redundant
 - iii. Clearly explained in terms of their value to the user's inquiry
- 8. Handling Unknown or Uncertain Information**
- a) If the agent cannot confidently identify the correct standard or section:
 - i. Do not guess
 - ii. Provide a courteous statement acknowledging the uncertainty
 - iii. Direct the user to Standards@ies.org for authoritative assistance
 - b) Example: *I am unable to determine the appropriate standard for this topic. For definitive guidance, please contact Standards@ies.org.*

- 9. Tone and Style Requirements**

- a) Maintain a tone that is:
 - i. Professional
 - ii. Neutral
 - iii. Academic
 - iv. Respectful
 - b) Avoid conversational filler, speculation, or personal opinions.
 - c) Keep responses concise while ensuring completeness and clarity.
- 10. Prohibited Behaviors**
- a) The agent must not:
 - i. Provide health or safety guidance
 - ii. Offer design calculations, compliance determinations, or engineering judgments
 - iii. Recommend actions that could be interpreted as legal, financial, safety or contractual advice
 - iv. Infer or fabricate content from standards not explicitly referenced

- 11. Predefined responses**

- a) If user requests lighting recommendations for an application that is not covered in current IES Standards
 - i. Respond similarly to: *There may not be explicit lighting recommendations for that application within the current body of IES Standards. Please review the monthly IES Ignite Newsletter for upcoming public review periods and publications. Similar applications include _____.* Would you like IES recommendations for any of those applications?
- b) If a user asks when a new or revised standard will be published
 - a) Respond similarly to: *Please review the monthly IES Ignite Newsletter for upcoming public review periods and publications.*

- **Summary of sources to be indexed**

- Current IES Standards (perhaps staff OneDrive folder...)
 - Metadata (from Vitrium? Discuss): title, description, author (technical committee), etc
- *Deprecated IES Standards (perhaps staff OneDrive folder...)
 - **Only* for the purpose of answering: "what changed" or "what is new/removed" in the current standard, or similar
 - Deprecated standards shall never be referenced/cited to answer any other form of question.
 - *If we ever license out API, this should not be provided for external use*
- IES.org resources, including:
 - Events calendar
 - eLearning courses
 - LC Study Group registration/schedule
 - Leukos
 - Upcoming webinars, virtual symposia
 - [LS-1 Nomenclature](#) (currently a separate webpage, but may be integrated into Vitrium in future)
 - [Standards Toolbox](#)
- IES Section Websites
 - Upcoming events
 - *If we ever license out API, this should never be provided for external use*
- Wicket, including:
 - User 'Section' affiliation
 - User LinkedIn/Website
 - Active technical committees
 - Active members (to be cross-referenced with "author" metadata from standards, which should indicate the authoring Technical Committee)
 - *If we ever license out API, this should never be provided for external use*

EXAMPLE SEARCH #1

Indexed From / Source

Natural-language AI Agent to help user narrow down their search.

Professional tone.

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IES StandardSearch

What would you like to look up? Type “what can I do” for examples of the types of questions you can ask.

What can I do?

Here are a few examples of the types of questions you can ask:

- What are the IES lighting recommendations for library stacks?
- I have a list of 10 lighting applications. What are the IES lighting recommendations for each?
- Which standard contains IES lighting definitions?
- What are some of the revisions in the current ANSI/IES RP-6?
- Where can I find relevant IES recommendations on lighting outdoor spaces?
- Which documents are authored by the IES Testing Procedures Committee?
- How does vision change as people age?

What are the IES lighting recommendations for the following applications:

Office kitchen
Office hallways
Office front façade
Office entrance
Office parking lot
Office meeting room
Restrooms
Copy Room
Desks

Below are a variety of resources from IES Standards that may be applicable to your search.

NOTE TO DEVELOPERS

Initial release: User may paste a *list* of applications into the chatbot to return a string of illuminance table results.

Future Development: User can upload Excel file (Room Schedule, OPR, etc) for a project, to lookup multiple illuminance table results simultaneously. The tool may be able to ‘plug in’ data from IES standards into their table, for recommended illuminance levels etc.

Discuss

Future Development: User uploads ‘master specs’ and can request improvements/suggestions based on IES Standards. E.g. “Improve my master specs and add TM-30 guidance for color rendition specifications”

Discuss

View Results: Office parking lot

View Results: Office kitchen

View Results: Office hallways

View Results: Office front facade

View Results: Office entrance

View Results: Office meeting room

Below are a few resources related to “office meeting rooms.” Begin a new search to dig deeper or open the recommended standards for full context.

Illuminance Level Recommendations

Recommended Maintained Illuminance Targets ^(a, b)											
Veiling Reflection Risk Light Level for Task or Area?		TS = Task Surface: Recommended Illuminances are at height of task surface above finished floor (AFF)									
		Horizontal (E _h)					Vertical (E _v)				
		Target E _h @ Height AFF		Uniformity Ratio			Target E _v @ Height AFF		Uniformity Ratio		
		C	Max	Avg	Ratio	Basis	C	Max	Avg	Ratio	Basis
APPLICATION TASK/AREA	Task High or Med Area	T	Lux @ m	(Fc @ Ft)	Min	Ratio	T	Lux @ m	(Fc @ Ft)	Min	Ratio
	Low Area	T	Lux @ m	(Fc @ Ft)	Min	Ratio	T	Lux @ m	(Fc @ Ft)	Min	Ratio
INTERIORS - COMMON APPLICATIONS											
Meeting ³											
Discourse	A	P	300 @ 0.76	(30 @ 2.5)	Avg		M	100 @ 1.22	(10 @ 4.0)	Avg	
Presentation surfaces ⁴	T	Q	400 @ 0.76	(40 @ 2.5)	Avg		N	150 @ 1.22	(15 @ 4.0)	Avg	

Results from IES Standards

Open ANSI/IES RP-1-24 Recommended Practice: Lighting Office Spaces

6.3 Conference and Meeting Rooms

Visual tasks in conference and meeting rooms range from casual to challenging, and from horizontal to vertical surfaces. Therefore, **two or more lighting systems should be planned to provide flexibility**, including:

- A general lighting system in which control is provided by switching or dimming to vary the illuminance
- A supplementary downlighting system with a dimmer control for image projection and other low level illumination requirements; due to improved technology and the reduced cost of dimming systems, it is sometimes effective to incorporate dimming into a general ambient system, eliminating the need for a second downlighting system.
- A perimeter or wall-wash lighting system controlled with dimmers for better visual appeal, particularly where presentation materials may be wall mounted
- A pre-set lighting control system to provide for easy and consistent operation of the lighting as tasks or functions change

Open ANSI/IES RP-10-20+E2 Recommended Practice: Common Applications

3.5 Conferencing

Conferencing runs the gamut from very simple single purpose functionality (for example, meet to discuss; see Figure 3-5) to rather complex multipurpose tasks (for example, meet to discuss, present, strategize, and learn, or video conference with others). The lighting is influenced accordingly, and by the formality of the setting. Where presentations are common, such as a board room or a meeting room, and where the participants might number two dozen, the lighting should help define the formality and procedural aspects of the meeting. Accenting on presenter positions is appropriate. Some conference settings strike a balance. In these situations, presentations are common but intended to stimulate a group discussion and learning situation. Here, although presentation surfaces are still accented, presenters might not be specially illuminated.

Auto-generated response based on IES Standards

This section is for informational purposes only. It contains an AI-generated response derived from current IES standards and may contain errors or omissions. It has not been edited by subject matter experts and is not intended as professional, legal, or financial advice. Refer to the body of IES Standards. This text may not be copied, reproduced nor distributed.

Acknowledge. Click to view.

IES recommends that office meeting rooms be lit to support clear facial visibility, comfortable reading, and effective presentations. The primary guidance comes from ANSI/IES RP-1-20: Lighting Office Spaces, which defines illuminance targets and supporting design considerations for conference and meeting rooms.

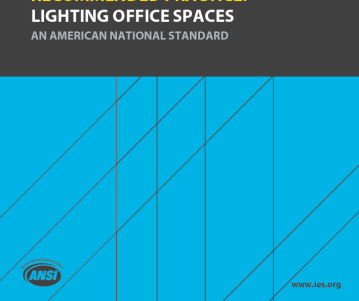
RP-1-20 provides broader design guidance for office spaces, including meeting rooms. Key principles include:

- ✓ Balanced Ambient + Task Lighting
- Ambient lighting should support comfortable visibility without glare.
- Supplemental vertical lighting improves facial recognition and communication.
- ✓ Glare Control
- Use luminaires with good shielding and low UGR (Unified Glare Rating).
- Avoid direct light paths toward presentation screens.
- ✓ Vertical Illuminance Matters
- Meeting rooms rely heavily on face-to-face communication.
- Vertical illuminance on faces and walls improves visual comfort and engagement.
- ✓ Flexible Controls
- Dimming is essential to accommodate presentations, video conferencing, and collaborative work.
- Multiple scenes (e.g., “Discussion,” “Presentation,” “Video Call”) are recommended.
- ✓ Color Quality
- Use high-quality color rendering (TM-30 guidance is often applied in practice).
- Consistent CCT (typically 3000–4000 K) helps maintain a professional environment.

Collapse

Recommended IES Standards

Open



ANSI/IES RP-1-24 Recommended Practice: Lighting Office Spaces

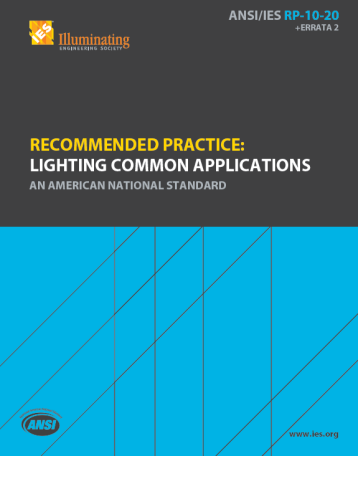
Subscribe 7-day Loan 5-yr PDF Compare access options

This Recommended Practice summarizes the photobiological hazards of exposure to ultraviolet (UV) radiation and to provide recommendations to minimize the risks of such effects from ultraviolet lamp systems. Target applications include consumer, industrial, scientific and medical applications for optical radiation emitted between 180 nm and 3,000 nm and in the spectral region of 180 nm to 400 nm.

Authored by the IES Education, Library and Office Spaces Committee

Chair, [Jane Doe](#) (links to website or LinkedIn). Vice Chair, [Jane Doe](#). Members: [John Doe](#), [Jane Doe](#), [John Doe](#), [Jane Doe](#), [John Doe](#).

Open



ANSI/IES RP-10-20 Recommended Practice: Lighting Common Applications

Subscribe 7-day Loan 5-yr PDF Compare access options

This Recommended Practice primarily addresses design considerations and illuminance criteria for common areas, which should influence luminaire optical selections, light source choices, and final layouts. Some of these applications, such as lobbies, make first impressions. Others, like conferencing and food service, can be the amenities that set one employer apart from others. Lighting for these applications is important and can infuse a facility with uncommon character. Daylighting can be quite effective in addressing illuminances in many of these applications.

Authored by the IES Education, Library and Office Spaces Committee

Chair, [Jane Doe](#) (links to website or LinkedIn). Vice Chair, [Jane Doe](#). Members: [John Doe](#), [Jane Doe](#), [John Doe](#), [Jane Doe](#), [John Doe](#).

View Results: Restrooms

View Results: Copy room

View Results: Desks

Recommended IES Content (Education, Research & Events)



Connect with your IES Section

You are part of the IES [city] Section [link to section website]. Connect with your local section.

Upcoming Events:

MARCH 26, 2026 | 6:00 pm - 6:00 pm EDT

IES Detroit March EP Event

Full Throttle Adventure Park 42223 W 12 Mile Rd, Novi, MI, United States

Dominate the adventure race with a non-stop 2-hour blast of High-Speed Coasting, Steady State and Virtual Reality Fun at Full Throttle Adventure Park in Novi. Not only will there be...

Did you know the following authors are also members of your IES Section?

IES Education, Library and Office Spaces Committee

- John Doe (links to LinkedIn if in Wicket, or to Contributor Page...?)
- Mary Doe

IES Light + Design Committee

- John Doe (links to LinkedIn if in Wicket, or to Contributor Page...?)
- Mary Doe

IES StandardSearch

What would you like to look up? Type “what can I do” for examples of the types of questions you can ask.

What can I do?

Here are a few examples of the types of questions you can ask:

- What are the IES lighting recommendations for library stacks?
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- Which documents are authored by the IES Testing Procedures Committee?
- How does vision change as people age?

Where can I find IES lighting definitions?

Below are a variety of resources from IES Standards that may be applicable to your search.

Results from IES Standards

Open ANSI/IES LS-1-25 Lighting Science: Nomenclature and Definitions for Illuminating Engineering

The IES defines illumination engineering terms in ANSI/IES LS-1-25, is available here online. You can search terms by their first letter using the categories listed in the Categories window, by the first letter of the term in the alphabetical list below, or by typing the term in the search window. You can find the Introduction material for LS-1-25 on these pages:

- Copyright, Disclaimer, and American National Standards Statement
- Preface, Nomenclature Committee members, Introduction, and Scope
- Spectral Parameters Used in Illumination Engineering

Open ANSI/IES LM-11-20(R2023) Approved Method: Photometric Testing of Searchlights Using Incandescent or HID Sources

The terms used in this document follow the definitions given in Normative Reference 2.1. Additional terms are defined below.

3.1 auxiliary equipment

The ballast and the igniter needed for the proper operation of discharge lamps.

3.2 units

The units of electrical measurement used in this Approved Method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of the CIE recommended systems.

Open ANSI/IES LM-31-20(R2023) Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires Using Incandescent Filament or High Intensity Discharge Lamps

The terms used in this document follow the definitions given in Normative Reference 2.5. Additional terms are defined below.

3.1 auxiliary equipment

The auxiliary equipment includes the ballast, capacitor, and the igniter needed for the proper operation of discharge lamps.

3.2 area lighting

In this document, area lighting refers to exterior illumination provided by luminaires (such as outdoor wall packs, bollards, and general area lighting luminaires) that may not be designed to fit into the standard Type I, II, III, IV, V, or VS classifications.

3.3 pre-burning

Pre-burning is defined as operating a previously seasoned lamp for a sufficient period of time such that it reaches thermal equilibrium, the mercury locates at the coolest zone of the lamp, and a stable light output is achieved under the test conditions.

3.4 test distance

The test distance shall be defined as the distance along the light path (direct and/or reflected) from the center of rotation of the goniosphotometer axes as shown in Figure 6-2 (Section 6.2), to the surface of the photodetector.

3.5 units

The units of electrical measurement used in this approved method are the volt, ampere, and watt. The units of photometric measurement are the lumen and the candela. Color is specified using chromaticity via chromaticity coordinates. 1976 (u, v) chromaticity coordinates are preferred, but 1931 (x, y) chromaticity coordinates may also be used.

Open ANSI/IES LM-35-20(R2023) Approved Method: Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps

The terms used in this document follow the definitions given in Normative Reference 2.1. Additional terms are defined in the subsections below.

3.1 units

The units of electrical measurement used in this Approved Method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of the CIE recommended systems.

3.2 auxiliary equipment

The ballast and the igniter needed for the proper operation of discharge lamps.

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The official source for IES lighting definitions is ANSI/IES LS-1, the Illuminating Engineering Society's authoritative, open-access glossary of lighting science terminology. It is the single, continuously maintained standard that defines all formal lighting terms used across IES documents.

Where to Find IES Lighting Definitions

- 1. ANSI/IES LS-1-25 -- Lighting Science: Nomenclature and Definitions for Illuminating Engineering

This is the official IES definitions standard, freely accessible online.

It contains 1,300+ lighting terms, including photometry, color science, vision, equipment, and application-specific terminology.

You can browse alphabetically (A-Z) or search directly.

What you'll find inside:

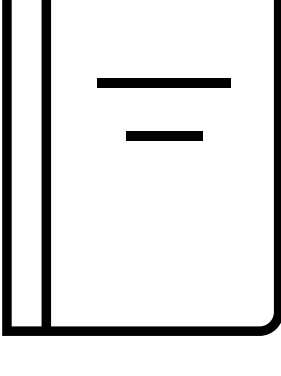
- Formal definitions for all lighting engineering terms
- Spectral and photometric terminology
- Vision science concepts
- Equipment and system definitions
- Terms used across all IES standards

This is the definitive reference used by lighting professionals, specifiers, and standards developers.

Collapse

Recommended IES Standards

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ANSI/IES LS-1-25 Lighting Science: Nomenclature and Definitions for Illuminating Engineering

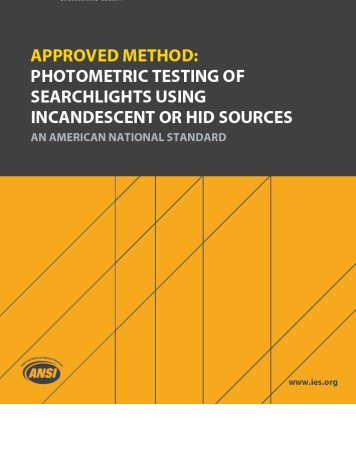
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Illuminating engineering, strictly speaking, comprises the production, measurement and application of light, or radiant energy within the limits of the visual spectrum. Since, however, most light sources furnish radiant energy also in the adjoining infrared and ultraviolet regions, and since many lamp-type devices are used for the production of radiant energy in these regions, it is customary to include the infrared and ultraviolet within the province of the illuminating engineer.

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ANSI/IES LM-11-20(R2023) Approved Method: Photometric Testing of Searchlights Using Incandescent or HID Sources

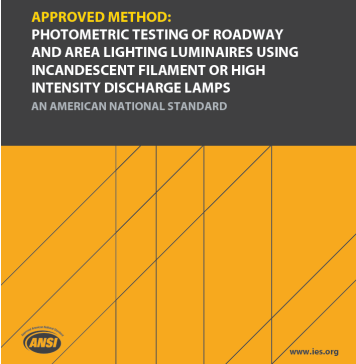
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This Recommended Practice primarily addresses design considerations and illuminance criteria for common areas, which should influence luminaire optical selections, light source choices, and final layouts. Some of these applications, such as lobbies, make first impressions. Others, like conferencing and food service, can be the amenities that set one employer apart from others. Lighting for these applications is important and can infuse a facility with uncommon character. Daylighting can be quite effective in addressing illuminances in many of these applications.

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ANSI/IES LM-31-20(R2023) Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires Using Incandescent Filament or High Intensity Discharge Lamps

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This Lighting Measurement (LM) guide defines adequate and uniform methods for measuring and reporting the photometric characteristics of roadway and area lighting luminaires using incandescent filament or high intensity discharge lamps. It describes characteristics of luminaires and some components, as well as the requirements for the thermal environment and proper control of the electrical and mechanical systems involved. General test conditions and the testing procedure best suited for achieving accurate and consistent photometric results are defined. This guide does not cover testing of luminaires using low pressure sodium lamps.

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ANSI/IES LM-35-20(R2023) Approved Method: Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps

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This Approved Method applies to luminaires having a total field angle greater than or equal to 10 degrees. The method applies to luminaires that utilize a variety of light sources, including tungsten filament, tungsten-halogen, and high intensity discharge (HID) lamps. It does not apply to luminaires using low pressure sodium (LPS), fluorescent, or solid state lighting (SSL) sources. (For luminaires where the total field spread is less than 10 degrees, refer to ANSI/IES LM-11-20. Photometric testing of reflector-type lamps is described in ANSI/IES LM-20-20.)

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Here are a few examples of the types of questions you can ask:

- What are the IES lighting recommendations for library stacks?
- I have a list of 10 lighting applications. What are the IES lighting recommendations for each?
- Which standard contains IES lighting definitions?
- What are some of the revisions in the current ANSI/IES RP-6?
- Where can I find relevant IES recommendations on lighting outdoor spaces?
- Which documents are authored by the IES Testing Procedures Committee?
- How does vision change as people age?

What’s new in the current version of ANSI/IES RP-6?

The following lists several of the additions and revisions to ANSI/IES RP-6-24 *Recommended Practice: Lighting for Sports and Recreational Areas*.

NOTE TO DEVELOPERS

Deprecated Standards should *only* be referenced in *this type* of search, where a user is comparing two versions of IES standards to each other.

Notes:

- Only the *current* standard should be recommended for further reading
- Tone: natural-language responses should emphasize that the “[old standard] is deprecated and has been replaced by the [current standard].”
- Below, paste excerpts of sections “REVISED” or “ADDED.” Do not print any sections “removed” except in the opt-in natural-language response box below.

Results from IES Standards

Open ANSI/IES RP-6-24 *Lighting Applications: Lighting for Sports and Recreational Areas*

6.20 Pickleball (ADDED)

Pickleball is a fast multi-directional, aerial sport (see Figure 6-31). Players are required to react quickly and move with confidence. Regardless of the level of play, pickleball is frequently played with smaller audience sizes due to the size of the court and viewing opportunities. Lighting should, therefore, be designed based on the level of play to allow players to perform at capability.

The game is primarily played in three zones:

- **The non-volley zone**, which is immediately adjacent to the net, is where opposing players are only 4.3 m (14 feet) apart and where a struck ball can achieve speeds up to 130 kilometers per hour (about 80 miles per hour).
- **The transition zone**, which is between the nonvolley zone and the baseline, is where opposing players are approximately 6 to 7.6 meters (20 to 25 feet) apart.
- **The baseline**, which is where opposing players are the farthest apart, at 13.4 meters (44 feet) apart, and where ground strokes, similar to those in tennis, are used regardless of the level of play, pickleball is frequently played with smaller audience sizes due to the size of the court and viewing opportunities. Lighting should, therefore, be designed based on the level of play to allow players to perform at capability.

Open ANSI/IES RP-6-24 *Lighting Applications: Lighting for Sports and Recreational Areas*

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- **The transition zone**, which is between the nonvolley zone and the baseline, is where opposing players are approximately 6 to 7.6 meters (20 to 25 feet) apart.
- **The baseline**, which is where opposing players are the farthest apart, at 13.4 meters (44 feet) apart, and where ground strokes, similar to those in tennis, are used regardless of the level of play, pickleball is frequently played with smaller audience sizes due to the size of the court and viewing opportunities. Lighting should, therefore, be designed based on the level of play to allow players to perform at capability.

Open ANSI/IES RP-6-24 *Lighting Applications: Lighting for Sports and Recreational Areas*

3.5.3 Light Sources (REVISED)

LEDs are currently the most common light source used in sports lighting. Other sources include metal halide, fluorescent, and plasma lighting. All are viable, but LEDs have the most advantages. Proper light source selection includes consideration of the particular requirements of the installation, economic factors, and the preferences of both designer and client.

A brief overview is provided in this section. Additional information about light sources may be found in ANSI/IES LP-4-20, *Lighting Practice: Electric Light Sources – Properties, Selection, and Specification* (see Preface).

LED luminaires have a high lumen maintenance factor and provide good color rendition. For these and other reasons, they have become the most viable solution for the entire range of sports lighting applications.

Open ANSI/IES RP-6-24 *Lighting Applications: Lighting for Sports and Recreational Areas*

5.11 Futsal (ADDED)

Futsal is a modified form of soccer that is played with five players per team (one of which is a goalkeeper). It is typically played indoors but can be played outdoors on a hardcourt like basketball (see Figure 5-9). Futsal is also played with a smaller ball with less bounce than a regular soccer ball. Futsal playing areas’ size can vary, as most fields are part of multi-use facilities or converted from another sport (see Figure 5-9). Due to the use of a smaller ball with less bounce, ball speeds are slower than in soccer. Light levels for futsal should be similar to those for soccer field levels. (Refer to the Annex A recommendations for soccer.)

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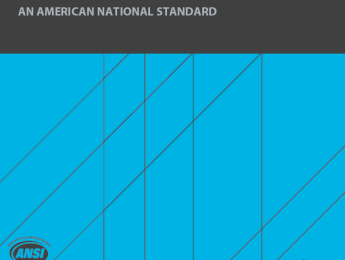
The newest RP-6 introduces major updates across environmental principles, color guidance, sport coverage, and lighting design methods. Below is a structured breakdown of all documented changes.

1. Integration of the IES-DSI Five Principles for Outdoor Lighting
RP-6-24 now embeds the Five Principles for Responsible Outdoor Lighting, aligning sports lighting with modern environmental and ecological best practices.
2. Updated Color Temperature Recommendations
The new edition revises CCT guidance, reflecting:
 - Reduced reliance on high-CCT sources
 - Better alignment with environmental concerns
 - Improved visual comfort and broadcast needs
3. New Guidance for Special Event Lighting
RP-6-24 adds a dedicated section on temporary and special-event lighting, addressing:
 - Portable lighting systems
 - Safety considerations
 - Temporary uniformity and glare control
4. Stairway & Aisle Lighting Requirements Added
The standard now includes explicit recommendations for spectator circulation areas, improving safety and consistency across venues.
5. Removal of Legacy Light Sources
All references to outdated lamp technologies have been removed, reflecting the industry’s full transition to LED and advanced optical systems.
6. Clarified Classification of Play Levels
RP-6-24 refines how sports are categorized by level of play, improving consistency in illuminance selection.
7. New & Updated Sports Coverage
The standard expands and updates its sport-specific guidance:
 - Added
 - Futsal
 - Pickleball (enhanced guidance)
 - Running tracks (enhanced guidance)
 - Removed
 - Dog racing
8. Expanded Annexes & Technical Methods
The 2024 edition includes updated annexes covering:
 - Computer calculations
 - Floodlight aiming methods
 - Field measurements
 - Uniformity metrics
 - Broadcast lighting considerations

----- Collapse -----

Recommended IES Standards

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ANSI/IES RP-6-22 *Recommended Practice: Lighting Sports and Recreational Areas*

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The purpose of this Recommended Practice (RP) is to provide the reader with recommendations to aid in the design of sports lighting systems. Popular sports such as baseball, tennis, basketball and football, as well as recreational social activities such as horseshoe pitching and croquet are covered. Venues for spectators of amateur, collegiate, and professional sports are complex facilities that should provide not only for the spectators but also the equipment used in modern sports broadcasting. This document does not address the needs of broadcasting; for this, the reader should look for guidance from the sports league or the project consultant.

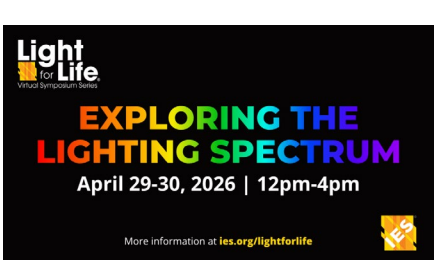
Authored by the IES Sports & Recreational Area Lighting Committee

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Recommended IES Content (Education, Research & Events)



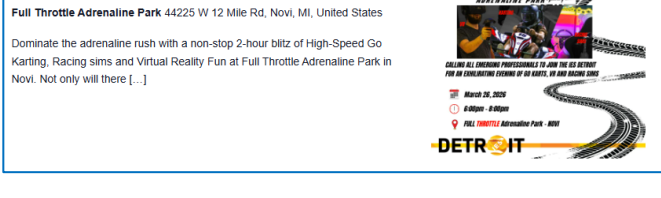
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- Where can I find relevant IES recommendations on lighting outdoor spaces?
- Which documents are authored by the IES Testing Procedures Committee?
- How does vision change as people age?

Where can I find relevant IES recommendations on lighting outdoor spaces?

What types of outdoor spaces would you like to look up? Responses such as parks, roadways, facades, or industrial areas will help narrow the search.

Exterior restaurants

The following lists several IES sources for recommendation on lighting exterior restaurants.

Illuminance Level Recommendations

OpenANSI/IES RP-43-25 Recommended Practice: Lighting Design for Outdoor Pedestrian Applications

LIGHTING FOR PEDESTRIANS IN OUTDOOR SPACES Recommended Average Maintained Illuminance Targets ⁽¹⁾											
TS = Task Surface; Recommended Illuminances are at height of task surface above finished floor (AFF)											
Application Task/Area	Horizontal (E _h)				Vertical (E _v)				Environmental and Visual Considerations		
	Target E _h @ Height AFF	Uniformity	Target E _v @ Height AFF	Uniformity	Glare (max)	Uplight (max)	Controls	Spectrum			
	Lux@2m	Fc@Ft	Ratio	Ratio Basis	Lux@2m	Fc@Ft	Ratio	Ratio Basis	Glare (max)	Uplight (max)	Controls
ATMOSPHERE AND ENJOYMENT Table A-1											
Outdoor Restaurants and Dining Areas (with full public accommodation)											
L24											
Lower limit (avg.)	20@0	2@0	8:1	Avg Min					5%	500 lm	2400 K to 4000 K
Upper limit (avg.)	100@0	10@0	8:1	Avg Min					5%	500 lm	2400 K to 4000 K
L23											
Lower limit (avg.)	20@0	2@0	8:1	Avg Min					5%	500 lm	2400 K to 3000 K
Upper limit (avg.)	80@0	8@0	8:1	Avg Min					5%	500 lm	2400 K to 3000 K
L22											
Lower limit (avg.)	10@0	1@0	10:1	Avg Min					5%	50 lm	2400 K to 3000 K
Upper limit (avg.)	40@0	4@0	10:1	Avg Min					5%	50 lm	2400 K to 3000 K
L21											
Lower limit (avg.)	5@0	0.5@0	20:1	Avg Min					3%	10 lm	2400 K to 3000 K
Upper limit (avg.)	10@0	1@0	20:1	Avg Min					3%	10 lm	2400 K to 3000 K
L20											
Lower limit (avg.)											
Upper limit (avg.)											
5.	Glare (max): Using the luminaire classification system (LCS) found in ANSI/IES TM-15, a maximum percentage of light emission just below horizontal (FVH) is recommended for luminaires emitting greater than 2,000 lumens. To be useful and beneficial, light will need to be targeted for the task and avoid creating potential glare or wasted emissions. See Section A.2.2 for additional guidance.										
6.	Uplight (max): Light emission limits are recommended for uplight, as measured by the total number of lumens emitted within the Uplight Low-Angle (UL) zone of the TM-15 Luminaire Classification System. See Section A.2.2 for additional guidance.										
7.	Controls: Directing light to hit its target is not sufficient to minimize ecological impacts on nocturnal species and habitat. Every effort should be made to reduce light when it is not needed. Extinguish or dim the light when it is not needed during times of vacancy, curfew, or seasonal conditions. See Section A.2.2 for additional guidance.										
8.	Controls: Illuminance reductions during vacancy, curfew, and nighttime hours may not be recommended when there are pedestrian safety and/or security reasons. See Section A.2.2 for additional guidance.										
9.	Controls: Using controls to dim or turn off lighting during vacancy, curfew, and seasonal conditions is represented by a percentage of full light output. For example, “20%” refers to 20% of full output. See Section A.2.2 for additional guidance.										
10.	Spectrum: There is not yet widespread agreement on a more relevant metric than CCT for outdoor spectral considerations. For this reason, CCT is used here. CCT values available for consideration are: CCT ≤ 1800 K, CCT ≤ 2400 K, CCT ≤ 3000 K, and CCT ≤ 4000 K. See Section A.2.2 for additional guidance.										

OpenANSI/IES RP-43-25 Recommended Practice: Lighting Design for Outdoor Pedestrian Applications

8.8.2 Outdoor Dining.

Outdoor dining is a broad term meant to represent areas within the outdoor space where the atmosphere or character identity attracts people to come and experience extended moments of enjoyment that typically include other people and the opportunity to eat or drink.

The lighting design process may begin by determining the lighting zone and confirming who the “landlord” of the dining area will be. These two answers will help the lighting designer understand ambient lighting conditions, adjacent lighting conditions, and desired branding for the dining atmosphere. Of note, outdoor dining areas may be found in L21 or L22 spaces, such as a municipal park, as well as L23 or L24 spaces, such as a pedestrian mall or public streetscape. For example, if the subject area is in a municipal park, then statutory lighting requirements might be the logical starting point for the lighting design. However, if the outdoor dining is at the edge of the public realm and serviced by private restaurants or retail food outlets, the character of the venue may be a more appropriate baseline at which to begin the design process.

OpenANSI/IES RP-43-25 Recommended Practice: Lighting Design for Outdoor Pedestrian Applications

8.8.2 Outdoor Dining.

Outdoor dining areas are often adjacent to architecture and other parts of the built environment, meaning that there are likely other light sources nearby, and transitional lighting techniques may be necessary to facilitate the change of primary task from navigation to relaxation. The challenge for the lighting designer is to create a luminous environment that is attractive and comfortable while also meeting lighting criteria related to personal navigation and reassurance. Whether the selected illuminance level on the dining table is 5 lux or 100 lux, the designer is advised to consider the overall lit environment as a lighting hierarchy. Consideration of Richard Kelly’s lighting design principles of “ambient luminescence,” “focal glow,” and “play of brilliants” is recommended.⁸⁷ In all cases, the lighting for outdoor dining areas will need to consider the lighting zone and ambient surroundings to support the foundational pedestrian tasks of orientation, navigation, and reassurance, in addition to creating the atmosphere and enjoyment tasks associated with an outdoor dining area.

Table A-3 in Annex A provides illuminance recommendations for outdoor dining areas based on ambient lighting conditions expected from community planning and the use of lighting zones. The specific illuminance target criteria present an acceptable range of illumination, allowing the designer to increase or decrease the light levels based on additional factors. For example, lower light levels may be desired for dining areas with low occupancy to create an intimate atmosphere, as compared with dining areas with higher patron traffic that need to be on the higher end of the range for navigation safety and reassurance.

OpenANSI/IES RP-9-23 Recommended Practice: Lighting Hospitality Spaces

12.2 Design Criteria and Solutions

In general dining spaces, supplemental systems, switching, and dimmers make the same space comfortable for breakfast, lunch, and dinner. Designers should consider the need to vary the quantity and quality of light to control light levels at different times of the day. Another consideration is that higher light levels are needed for cleanup. Automatic preset, programmable dimmers with variable fade rates are preferable to switches because they provide a smooth transition between levels that should be imperceptible to guests.

Food displays should be lighted so that they draw customers’ attention and allow them to clearly see the details of the food. Color rendering is more important for fresh foods than for packaged foods. A good general rule is that the illuminance used in food displays should be at least twice as high as that of surrounding areas. Designers should also consider the amount of heat emitted from luminaires that are used to highlight fresh, cooled, or frozen foods, although this now tends to be less of a concern when LED luminaires are used, as there is very little heat projected in the beam.

Depending on the type of restaurant and the intended clientele, lighting can vary the mood from subdued and relaxing to bright and lively. Dining spaces usually fall into three categories: intimate, leisure, and quick service.

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IES does not publish a single standard dedicated exclusively to “exterior restaurant lighting,” but the topic is well covered across several Recommended Practices that address hospitality, outdoor pedestrian environments, and façade/entrance lighting. The most authoritative sources are below.

1. ANSI/IES RP-9-23 — Lighting Hospitality Spaces
Primary reference for restaurants (interior and exterior zones). While RP-9-23 focuses heavily on interior hospitality environments, it includes guidance relevant to restaurant entries, outdoor dining areas, and hospitality-adjacent exterior spaces. Look specifically at:
Section 7 — Exterior Arrival, Entry, and Transition Zones
Discusses visual comfort, wayfinding, and luminance balance at hospitality venues.
Section 10 — Outdoor Dining & Social Spaces
Addresses ambience, glare control, and recommended illuminance ranges for exterior seating areas.
Recommended additional reading:
ANSI/IES RP-43-23 for exterior pedestrian environments
ANSI/IES LP-2-20 for lighting design process considerations

2. ANSI/IES RP-43-23 — Lighting for Outdoor Pedestrian Spaces
This is the most important standard for any exterior restaurant area used by pedestrians, including:
Outdoor dining terraces
Sidewalk cafés
Restaurant plazas
Walk-up service windows
Adjacent pedestrian circulation zones
Key sections to review:
Section 5 — Pedestrian Needs and Visual Tasks
Section 6 — Recommended Illuminance Values
Section 7 — Luminance, Glare, and Visual Comfort
Appendix A — Lighting Zones (L20–L24) for context-appropriate light levels
RP-43-23 provides the maintained illuminance targets, uniformity ratios, and design considerations most applicable to exterior restaurant environments.

Collapse

Recommended IES Standards

OpenANSI/IES RP-43-25 Recommended Practice: Lighting Design for Outdoor Pedestrian Applications

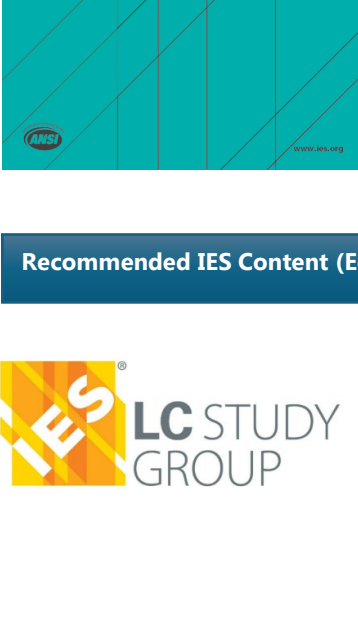


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The purpose of this Recommended Practice (RP) is to provide pedestrian-oriented illumination recommendations for the reassurance, safety, comfort, amenity, and enjoyment of people in outdoor environments in lighting zones L2-1 through L2-4. This RP includes recommendations beyond illuminance, which when considered alone is inadequate for addressing the visual needs of pedestrians. Rather, it takes a comprehensive approach and makes recommendations based on lighting zone, glare avoidance, spectrum, and other visually influential conditions. Application of these recommendations will ultimately enhance the visual experience for people, while also respecting the environment.

Authored by the IES Outdoor Nighttime Environments Committee
Current members: Chair, Jane Doe [links to website or LinkedIn]. Vice Chair, Jane Doe. Members: John Doe, Jane Doe, John Doe, Jane Doe, John Doe.

OpenANSI/IES RP-9-23 Recommended Practice: Lighting Hospitality Spaces



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Design of lighting systems for hotel applications. The intent of this document is to address how all of the various components of hospitality lighting need to be an integrated system of layers as well as uniquely represented areas of illumination.

Authored by the IES Sports & Recreational Area Lighting Committee
Chair, Jane Doe [links to website or LinkedIn]. Vice Chair, Jane Doe. Members: John Doe, Jane Doe, John Doe, Jane Doe, John Doe.

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Example of view when “Read More” is clicked for Illuminance Level Rec.

- One 'block' per table (typically 1 per standard, but if a standard has more than 1 table then may need >1 table for that standard)
- Include all footnotes from the entire table, whether referenced or not, to avoid transcription errors and complicated coding
- Include all General Notes from the entire table. This will be "Annex A."
- Include all References if possible. This might require some extra coding.

"Open" drives users to the page referenced in the relevant excerpt. For tables with multiple 'matches,' could either drive user to page of the specific row of text for each match (ideal), or to the first page of the relevant table (acceptable).

Illuminance Level Recommendations													
Application Task/Area	Class of Play	Horizontal (E _h)						Vertical (E _v)					
		Target E _h @ Height AFF			Uniformity			Target E _v @ Height AFF			Uniformity		
		C A T	Max Avg Lux@0m	Max Avg Min	CV	Ratio	Ratio Basis	C A T	Max Avg Lux@0m	Max Avg Min	CV	Ratio	Ratio Basis
Ice Hockey ANSI/IES RP-6-22 Table A-1													
Hockey rink	I	U	1500@0.91	Avg	0.13	1.5:1	Max:Min	0	0@0		0	0	
Hockey rink	II	T	1,000@0.91	Avg	0.17	1.7:1	Max:Min	0	0@0		0	0	
Hockey rink	III	S	750@0.91	Avg	0.21	2.5:1	Max:Min	0	0@0		0	0	
Hockey rink	IV	R	500@0.91	Avg	0.25	3:1	Max:Min	0	0@0		0	0	
Ice Skating (Speed*) ANSI/IES RP-6-22 Table A-1													
Skating area	III	P	300@0.00	Avg	0.21	2:1	Max:Min	0	0@0		0	0	
Skating area	III	O	200@0.00	Avg	0.25	3:1	Max:Min	0	0@0		0	0	
Skating area	IV	M	100@0.00	Avg	0.3	4:1	Max:Min	0	0@0		0	0	
Roller Skating ANSI/IES RP-6-22 Table A-1													
Skating rink	II	R	500@0.91	Avg	0.21	2.5:1	Max:Min	0	0@0		0	0	
Skating rink	III	P	300@0.91	Avg	0.25	3:1	Max:Min	0	0@0		0	0	
Skating rink	IV	O	200@0.91	Avg	0.3	4:1	Max:Min	0	0@0		0	0	
Figure Skating ANSI/IES RP-6-22 Table A-1													
Skating rink	I	-	1250@0.91	Avg	0.13	1.7:1	Max:Min	0	0@0		0	0	
Skating rink	II	T	1,000@0.91	Avg	0.21	2.5:1	Max:Min	0	0@0		0	0	
Skating rink	III	S	750@0.91	Avg	0.25	3:1	Max:Min	0	0@0		0	0	
Skating rink	IV	R	500@0.91	Avg	0.3	4:1	Max:Min	0	0@0		0	0	
GENERAL NOTES													

IES ILLUMINANCE RECOMMENDATIONS: SPORTS LIGHTING APPLICATIONS

General Notes:

These General Notes provide context and general guidance for the illuminance recommendations contained in this table, which includes illuminance criteria for tasks and areas in *ANSI/IES RP-6-20, Recommended Practice: Sports and Recreational Lighting*. For tasks not included in this table, the reader is referred to the appropriate *Commercial, Residential, and Industrial Applications* tables.

Maintained Illuminance Target values are consensus recommendations for minimum (Min), average (Avg), or maximum (Max) maintained illuminance levels for application tasks or areas described in the table at heights above finish floor (AFF, for interior applications) or above finish grade (AFG, for outdoor applications), for normally sighted people under 65 years of age. This table should be used in conjunction with other relevant design considerations following these governing criteria:

In cases where the height of the visual task may vary, the abbreviation TS (for "task surface") is used. The illuminance criteria then apply at the height of the visual task.

Light loss factors such as luminaire dirt depreciation, lumen depreciation, reduced room surface and furniture reflectances, and other design criteria should be used in calculating maintained illuminances. (Refer to *ANSI/IES LS-6-20, Lighting Practice: Calculation of Light and Its Effects*.)^[1] and *ANSI/IES NALMCO RP-36-20, Recommended Practice for Lighting Maintenance*^[10] for additional information.)

Illuminance levels are designated for Tasks or Areas. "T" (task) designates a visual task, and *illuminance level* refers to that of the localized task areas, such as a desk or reading surface (horizontal illuminance), or library stacks (vertical illuminance); the illuminance recommendations apply to that task area only, through the total contribution of general and task lighting. "A" (area) designates an area or room where the lighting criterion applies to an entire area under consideration, such as a corridor floor or the field of play in a soccer field. When multiple tasks are performed within the same space, tasks should be ranked by importance, prevalence, or frequency using data that may be available from the client. From that information, the most commonly occurring task with the highest recommended illuminance should govern the illuminance recommendation for the task or area. Localized task lighting can be used for instances when additional illumination is necessary for infrequent, but more visually demanding, tasks.

The letter code in the CAT column refers to the recommended illuminance categories found in Table A-2 of *ANSI/IES RP-10-20, Recommended Practice: Lighting Common Applications*.^[11]

Variances to Maintained Illuminance Target Values:

Health code and safety code requirements supersede these recommendations.

When safety and security or human-vehicular proximity are significant concerns, recommended values are to be minimum maintained illuminances for the task area. For more guidance in applications when security is a concern, refer to *IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure*.^[12]

An approximate lux-to-footcandle conversion factor of 10:1 is used in these tables, instead of the more accurate conversion factor of 10.76:1.^[13] Acceptable tolerances for lighting calculations during the design process are within ±10% of the target value. If a predicted value is below a target recommendation by more than 10%, then a significant percentage of the users of the system may not find the visibility acceptable. If a predicted value exceeds a target recommendation by more than 10%, then over-lighting and energy misuse may result.

When a majority of the occupants of a space are over 65, the illuminance recommendations should be doubled. For special considerations for seniors or visually impaired people of any age, refer to *ANSI/IES RP-28-20, Recommended Practice: Lighting and the Visual Environment for Older Adults and the Visually Impaired*.^[14] Localized additional task lighting should be considered for occupants who may require additional lighting, before selecting higher illuminance criteria for the entire space or group.

For visual tasks categorized as P through Y and under lighting conditions where the light source S/P Value is different from 1.0, variances are allowed in accordance with *ANSI/IES TM-24-20, Technical Memorandum: An Optional Method for Adjusting the Recommended Illuminance for Visually Demanding Tasks within IES Illuminance Categories P through Y Based on Light Source Spectrum*.^[15]

ANSI/IES RP-43-25 Recommended Practice: Lighting for Pedestrians in Outdoor Spaces

Application Task/Area	Horizontal (E _h)				Vertical (E _v)				Environmental and Visual Considerations			
	Target E _h @ Height AFF		Uniformity		Target E _v @ Height AFF		Uniformity					
	Lux@0m	Fc@Ft	Ratio	Ratio Basis	Lux@0m	Fc@Ft	Ratio	Ratio Basis	Glare (max)	Uplight (max)	Controls	Spectrum
Spectator Seating, Stairs, Ramps, Walkways (adjacent to field)												
L24												
Lower limit (avg.)	50@0	5@0	5:1	Avg:Min					5%	500 lm		2400 K to 4000 K
Upper limit (avg.)	100@0	10@0	5:1	Avg:Min					5%	500 lm		2400 K to 4000 K
L23												
Lower limit (avg.)	40@0	4@0	5:1	Avg:Min					5%	500 lm		2400 K to 4000 K
Upper limit (avg.)	80@0	8@0	5:1	Avg:Min					5%	500 lm		2400 K to 4000 K
L22												
Lower limit (avg.)	20@0	2@0	5:1	Avg:Min					5%	50 lm		2400 K to 4000 K
Upper limit (avg.)	40@0	4@0	5:1	Avg:Min					5%	50 lm		2400 K to 4000 K
L21												
Lower limit (avg.)	10@0	1@0	5:1	Avg:Min					3%	10 lm		2400 K to 4000 K
Upper limit (avg.)	20@0	2@0	5:1	Avg:Min					3%	10 lm		2400 K to 4000 K
L20												
Lower limit (avg.)												
Upper limit (avg.)												

- APPLICATION TASK/AREA NOTES
- Data fields with no value indicate there is no expectation of needing a recommendation for that application or task.
 - For average illuminance, a range is recommended instead of a single value. The values shown are the lower and upper ends of this average illuminance range. The choice of illuminance level within a range will vary according to the conditions of each site and task. Refer to [Table A-1](#) for guidance in selecting the illuminance target value.
 - If an area is environmentally sensitive, retroreflective materials should be considered before adding area lighting.
 - Outdoor lighting carries a responsibility to balance human and environmental needs. When human vision is challenged the most, the quality of the light provided is often more important than the quantity of light. Refer to [Section 5.0](#) for additional recommendations regarding responsible outdoor lighting.
 - Glare (max):** Using the luminaire classification system (LCS) found in *ANSI/IES TM-15*, a maximum percentage of light emission just below horizontal (FVH) is recommended for luminaires emitting greater than 2,000 lumens. To be useful and beneficial, light will need to be targeted for the task and avoid creating potential glare or wasted emissions. See [Section A.2.2](#) for additional guidance.
 - Uplight (max):** Light emission limits are recommended for uplight, as measured by the total number of lumens emitted within the Uplight Low-Angle (UL) zone of the TM-15 Luminaire Classification System. See [Section A.2.2](#) for additional guidance.
 - Controls:** Directing light to hit its target is not sufficient to minimize ecological impacts on nocturnal species and habitat. Every effort should be made to reduce light when it is not needed. Extinguish or dim the light when it is not needed during times of vacancy, curfew, or seasonal conditions. See [Section A.2.2](#) for additional guidance.
 - Controls:** Illuminance reductions during vacancy, curfew, and nighttime hours may not be recommended when there are pedestrian safety and/or security reasons. See [Section A.2.2](#) for additional guidance.
 - Controls:** Using controls to dim or turn off lighting during vacancy, curfew, and seasonal conditions is represented by a percentage of full light output. For example, "20%" refers to 20% of full output. See [Section A.2.2](#) for additional guidance.
 - Spectrum:** There is not yet widespread agreement on a more relevant metric than CCT for outdoor spectral considerations. For this reason, CCT is used here. CCT values available for consideration are: CCT ≤ 1800 K, CCT ≤ 2400 K, CCT ≤ 3000 K, and CCT ≤ 4000 K. See [Section A.2.2](#) for additional guidance.

GENERAL NOTES

IES ILLUMINANCE RECOMMENDATIONS: LIGHTING FOR PEDESTRIANS IN OUTDOOR SPACES

A.1 General Information Regarding the Illuminance Table

The illuminance recommendations found in this table are from *ANSI/IES RP-43-20, Recommended Practice: Lighting Exterior Applications, Table A-3* (in [Section A.3](#)), and are based on the lighting zone of the space or application. Lighting Zone 1 (LZ1), for example, is an area with a small human population and natural areas with flora and fauna that could be affected by nighttime lighting, and Lighting Zone 4 (LZ4) is a high-density urban area. Expectations for lighting are very different in these zones, since the levels of activity and the number of pedestrians will vary widely, and their visual systems will be adapted to low or high light levels, respectively. (Refer to *ANSI/IES RP-43-20, Section 2.3 Lighting Zones*, for LZ descriptions and examples.)

The applications and tasks are presented with a range of values the designer can select from, depending on other visual criteria (refer to [Section A.2 Illuminance Table Explanations and Adjustments](#), below). Energy considerations and lower lighting power density allowances may also require careful tailoring of illuminances to task needs.

For outdoor tasks not included in this table, the designer should choose a listed task that closely resembles the task in question (similar contrast and difficulty), or refer to the Commercial, Residential and Industrial Lighting Applications table.

Additional considerations:

Light loss factors such as luminaire dirt depreciation, lumen depreciation, reduced surface reflectance such as that from façades, and other design criteria should be used to adjust lighting calculations; these factors result in reduced lighting levels over time. The reader is referred to *ANSI/IES LS-6-20, Calculation of Light and Its Effects*^[1] and *ANSI/IES NALMCO RP-36-20, Recommended Practice: Lighting Maintenance*^[10] for additional information.

In cases where the height of the visual task may vary, the abbreviation TS (for "task surface") is used. The illuminance criteria then apply at the height of the visual task.

The values are consensus recommendations for normally sighted people under 65 years of age.

Illuminance targets are design goals; variations from them are expected and may be found at two stages of the construction process: at design time and at commissioning or occupancy time. Variances to maintained illuminance target values at design time include:

Health code and safety code requirements supersede these recommendations.

When safety and security or human-vehicular proximity are significant concerns, recommended values are to be minimum maintained illuminances for the task area. For more guidance in applications when security is a concern, refer to *IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure*.^[12]

An approximate lux-to-footcandle conversion factor of 10:1 is used in this table, instead of the more accurate conversion factor of 10.76:1.^[13] Acceptable tolerances for lighting calculations during the design process are within ±10% of the target value. If a predicted value is below a target recommendation by more than 10%, then a significant percentage of the users of the system may not find the visibility acceptable. If a predicted value exceeds a target recommendation by more than 10%, then over-lighting and energy misuse may result.

When a majority of the occupants of a space are over 65, the illuminance recommendations should be doubled. For special considerations for seniors or visually impaired people of any age, refer to *ANSI/IES RP-28-20, Recommended Practice: Lighting and the Visual Environment for Older Adults and the Visually Impaired*.^[14] Localized additional task lighting should be considered for occupants who may require additional lighting, before selecting higher illuminance criteria for the entire space or group.

A.2 Illuminance Table Explanations and Adjustments

This table includes recommendations based on human vision, visibility, and reassurance, as well as specific recommendations pertaining to lighting for responsible design. These considerations are explained below in [Sections A.2.1 and A.2.2](#) of these General Notes. Guidance for adjusting the target illuminances and related criteria is provided in [Section A.2.3](#).

Within this table, there are superscript numbers corresponding to the numbered notes at the bottom of that table.

A.2.1 Recommendations for Vision, Visibility, and Reassurance

These columns include:

A minimum and maximum range of average maintained horizontal illuminances, and recommended average-to-minimum uniformity ratios for those illuminances. If no specific task height is listed, the task plane is the ground. Illuminance targets and uniformity ratios are not intended to test the very extremes of the light pattern on a surface, but serve as a general guideline for uniformity. Guidance in choosing the low, middle, or high range of the illuminance zone is provided in *ANSI/IES RP-43-20, Section 5.2 How to Adjust Target Illuminances and Related Criteria*.

In some situations, the ground plane illuminance is not critical, and a target for vertical illuminance is listed instead; this illuminance is to be provided either on the task surface (such as a wall or façade) or at a specified face height. A minimum and maximum range of average maintained vertical illuminances is listed, along with recommended average-to-minimum uniformity ratios for those illuminances. In some cases, vertical illuminance is appropriate (such as the illumination of a building façade).

A.2.2 Recommendations for Responsible Design (Optical Control, Controls, Spectrum)

These columns include:

Optical Control: These columns include the Maximum Glare Rating ("G") and the Maximum Uplight Rating ("U"), respectively, from the *ANSI/IES TM-15-20 BUG* rating system for classifying light distribution from outdoor luminaires.^[15] G1 and U1 correspond to maximum classifications recommended in LZ1, G2 and U2 for LZ2, G3 and U3 for LZ3, and G4 and U4 for LZ4.

A G1 luminaire rating allows a much lower number of lumens emitted at angles close to horizontal, and G4 allows higher lumen output in that range, since a luminaire is perceived as less glaring if the viewer is adapted to a higher ambient light level. User expectations for lighting and glare are also generally higher in more populated areas with greater concerns for nighttime safety and security.

A U1 luminaire rating limits the number of upward lumens tightly, on the assumption that inherently dark rural and low-density areas also have more sensitive environmental areas, and that excess light that obscures the view of the night sky is unwanted by most residents. U2, U3, and U4 ratings correspond to increasingly more uplight allowed.

Note: The responsible design approach is to minimize uplight and glare whenever possible, even in urban areas. Therefore, although an LZ4 area can accommodate G4 and U4 luminaires according to the Optical Control columns, it is *strongly* recommended that G1 and U1 luminaires be implemented wherever possible on all projects. At the very least, utilitarian outdoor luminaires such as wall packs and parking lot luminaires should not emit more than a nominal number of lumens upward, and should also tightly limit glare.

Controls: This column recommends the minimum level of dimming appropriate for the application during curfew hours. This is for energy savings, reduced sky glow, and reduced destruction of natural habitat when areas are unused or during community curfew hours. In some cases, no controls reduction is recommended because it could create an unsafe condition. These cells are left blank.

Spectrum: Until there is a universally agreed upon metric for describing the short wavelength content of a light source (see *ANSI/IES RP-43-20, Section 3.5 Spectrum*), this table will refer to CCT. Therefore, the Spectrum guidance lists four categories: very high (VH): no CCT or melanopic DER^[16] restriction; high (H): CCT ≤ 4000 K (melanopic DER ≤ 0.6); medium (M): CCT ≤ 3000 K (melanopic DER ≤ 0.5); and low (L): CCT ≤ 2400 K (melanopic DER ≤ 0.3).

A.2.3 How to Adjust Target Illuminances and Related Criteria

This table lists target nighttime illuminance values and related criteria for outdoor public spaces. These values are based on in situ measurements and design experience by IES Lighting for Outdoor Public Spaces (LOPS) Committee members and advisors. The target light levels, uniformity ratios, and other criteria are guidelines based on the lighting zone of a project, acknowledging that criteria need to be different for areas with different characters and needs. Target illuminances are listed with a maximum and minimum value. There are reasons why the lighting professional may choose to target the high, middle, or low end of a range; many of these are listed below. If the reasons for the variance are documented for a project, it is acceptable to raise or lower light levels within that range. [Table A-1](#) provides weighting factors to assist the lighting professional. The factors (first column) are explained below in [Section A.2.1](#), and examples for their use are provided in [Section A.2.3.2](#).

Table A-1. Weighting Factors for Selecting Target Illuminance

Weighting Factors for Variance from Target Illuminance	Reduces Need for Light (-1)	No Impact on Need (0)	Increases Need for Light (1)
Usage levels	Low usage	Normal usage	High usage
Path priorities	Tertiary	Secondary	Primary
Seasonal changes	Winter snow		
Adjacencies	"Borrowed" light, glare concerns, or light unwanted	Neutral	Light wanted to improve perception of safety and activity; older pedestrians are a critical population
Client preferences; social settings; cultural expectations	Wants less light or prefers less uniformity	Neutral	Wants more light or prefers higher uniformity
Pavement condition	No level variation, high quality surface		Rough, uneven surface
Hazards or obstacles; wayfinding	Curbs, level changes, stairs, or obstacles have finishes that enhance contrast; or, retroreflective materials are used to create contrast		Level changes, dropping hazards, and obstacles do not use high-contrast materials
Mixed travel types			Bicycles, skateboards, or other non-motorized vehicles in addition to foot traffic
Safety; visibility of pedestrians	Subordinate to environmental issues	Neutral	Enhanced safety and/or visibility required
Glare from luminaires	Reduced glare for pedestrians, allowing clear visibility of pavement, objects, and surroundings		Higher glare for pedestrians, necessitating a higher ambient light level
Age of users	Users and supervisors predominantly less than 25 years old	Broad range of user ages	Users predominantly over 60 years old

A.2.3.1 Descriptions of the Weighting Factors

Usage levels. The number of users can increase or decrease the need for security in a public space. For example, if a large number of people are expected to attend a public concert, speech, or event, a higher light level may be needed to increase visibility for security. Conversely, if usage levels are expected to be low, the space is perceived as unsafe by the public, and there is no practical way to improve security, it may be prudent to reduce light levels or even eliminate electric lighting in order to discourage people from using that space or path at night.

Path priorities. Encouraging activity or foot-traffic in a primary outdoor space or path may be accomplished by raising light levels at night. Secondary paths may receive lower light levels, especially if there is a lower expectation of usage and crime.

Seasonal changes. Communities with winter snow cover may choose to reduce light levels during that season, since the snow reflection will increase target illuminances and, hence, visibility of objects, but will also redirect more light into the sky.

Adjacencies. The impact of lighting for a public space or path on adjacent buildings or uses needs to be considered.

Type of business or usage. People in residential units and hospitality accommodations may prefer reduced light trespass into bedroom windows. Conversely, if the adjacency is a retail business or community sports facility, more light from the public space may be welcome because it increases the appearance of activity and perception of safety.

Glare versus visibility. Improved visibility for security, aesthetic appearance, or nighttime activities may inadvertently introduce nuisance glare and light trespass to users on adjacent properties.

"Borrowed" light. Businesses or facilities that can be relied upon to have consistent shop window lighting or outdoor lighting may contribute sufficient direct or reflected light onto a public space such that additional lighting for the path or space may be reduced or eliminated. For example, some downtown areas can be lighted entirely by spill from shop windows or signage, if businesses agree to this.

Client preferences; social settings; cultural expectations. Facility owners, managers, and organizations may dictate higher or lower light levels for a variety of reasons, such as perceived safety, or an open or welcoming appearance. Nighttime outdoor seating in a high-end restaurant may be more appropriately lit with candles than electric lighting. Lower levels, spectral modifications, or glare control may be desired for reduced environmental impact or desired inconspicuousness. In either case, it is the duty of the lighting professional to communicate responsible lighting approaches. (For example: If a client requests lighting the ocean surf during turtle hatching season, the professional should voice concern about inappropriate light direction, quantity, and spectrum.)

Pavement condition. Uneven pavement or poorly maintained paving surfaces can pose a hazard that may necessitate higher light levels.

Hazards or obstacles; wayfinding. Curbs, walls, stairs, ramps, benches, and similar features can become hazards if not marked with high contrast materials or reflective surfaces. High contrast features can safely assist in wayfinding with minimal additional lighting. Retroreflective materials can be effectively used to create contrast or delineate edges when headlights or low-level luminaires are involved.

Reflectances and luminances of materials. Lighter color materials (>60% reflectance) reflect more light, potentially making objects more visible or increasing contrast between an object and its background. Lighter-colored materials used in paving result in a greater percentage of reflected light, thus contributing to sky glow, but they can also permit reducing target horizontal light levels because of the higher perceived brightness; very dark-colored materials (<10% reflectance) may necessitate higher light levels.

Mixed travel types. If bicycles or other non-motorized vehicles are sharing a path with pedestrians, it is more important to avoid collisions. If the path is narrow and bicycle speeds are high, higher light levels may be needed to increase visibility of users on foot.

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[1] An approximation of the CIE Melanopic Daylight (D65) Efficacy Ratio (melanopic DER) is included here for those specifiers familiar with it. (Ref: International Commission on Illumination (CIE), CIE S 026:2018, CIE System for Metrology of Optical Radiation for ipRGC-Influenced Responses to Light. Vienna: CIE; 2018.)

[11] Illuminating Engineering Society, ANSI/IES LS-6-20, Lighting Practice: Calculation of Light and Its Effects. New York: IES; 2020.

[12] Illuminating Engineering Society, ANSI/IES NALMCO RP-36-20, Recommended Practice: Lighting Maintenance. New York: IES; 2020.

[13] Illuminating Engineering Society, IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure. New York: IES; 2016.

[14] Houser K. Editorial: Footcandles or lux in IES publications? Leukos. 2019;15(4):249-50. DOI: 10.1080/15502724.2019.1638146.

[15] Illuminating Engineering Society, ANSI/IES RP-28-20, Recommended Practice: Lighting and the Visual Environment for Older Adults and the Visually Impaired. New York: IES; 2020.

[16] Illuminating Engineering Society, TM-15-20, Technical Memorandum: Luminaire Classification System for Outdoor Luminaires. New York: IES; 2020